

1. From (11.152)

$$E_{\parallel}(t) = -\frac{ze\gamma vt}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} \quad E_{\perp}(t) = \frac{ze\gamma b}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} \quad (1)$$

Thus the corresponding Fourier components are

$$\begin{aligned} E_{\parallel}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{-ze\gamma vt e^{i\omega t} dt}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} \quad \text{let } s \equiv \frac{b}{\gamma v} \\ &= \frac{-ze}{\sqrt{2\pi}} \frac{1}{(\gamma v)^2} \left[\overbrace{\int_{-\infty}^{\infty} \frac{t \cos \omega t dt}{(s^2 + t^2)^{3/2}}}^0 + i \int_{-\infty}^{\infty} \frac{t \sin \omega t dt}{(s^2 + t^2)^{3/2}} \right] \\ &= \frac{ize}{\sqrt{2\pi}} \frac{1}{(\gamma v)^2} \int_{-\infty}^{\infty} \frac{d}{dt} \left(\frac{1}{\sqrt{s^2 + t^2}} \right) \sin \omega t dt \\ &= -\frac{ize}{\sqrt{2\pi}} \frac{\omega}{(\gamma v)^2} \int_{-\infty}^{\infty} \frac{\cos \omega t}{\sqrt{s^2 + t^2}} dt \end{aligned} \quad (2)$$

$$E_{\perp}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{ze\gamma b e^{i\omega t} dt}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} = \frac{ze}{\sqrt{2\pi}} \frac{b}{\gamma^2 v^3} \int_{-\infty}^{\infty} \frac{\cos \omega t dt}{(s^2 + t^2)^{3/2}} \quad (3)$$

From [DLMF 10.32.E11](#) we have the integral representation of the Modified Bessel function

$$K_{\nu}(\omega s) = \frac{\Gamma\left(\nu + \frac{1}{2}\right)(2s)^{\nu}}{\sqrt{\pi}\omega^{\nu}} \int_0^{\infty} \frac{\cos \omega t dt}{(s^2 + t^2)^{\nu + \frac{1}{2}}} \quad (4)$$

turning (2) and (3) into

$$E_{\parallel}(\omega) = -\frac{ize}{\sqrt{2\pi}} \frac{\omega}{(\gamma v)^2} \cdot 2K_0(\omega s) \frac{\sqrt{\pi}}{\Gamma\left(\frac{1}{2}\right)} = -\frac{ize}{\gamma v b} \sqrt{\frac{2}{\pi}} \xi K_0(\xi) \quad \text{where } \xi \equiv \omega s = \frac{\omega b}{\gamma v} \quad (5)$$

$$E_{\perp}(\omega) = \frac{ze}{\sqrt{2\pi}} \frac{b}{\gamma^2 v^3} \cdot 2K_1(\xi) \frac{\sqrt{\pi}\omega}{\Gamma\left(\frac{3}{2}\right) \cdot 2s} = \frac{ze}{bv} \sqrt{\frac{2}{\pi}} \xi K_1(\xi) \quad (6)$$

2. From the result of problem 13.2, the energy transfer is

$$\Delta E = \frac{\pi e^2 |E(\omega_0)|^2}{m} = \frac{2z^2 e^4}{mb^2 v^2} \xi^2 \left[\frac{1}{\gamma^2} K_0^2(\xi) + K_1^2(\xi) \right] \quad (7)$$

The adiabatic condition (see paragraph around (13.8)) is represented by large $b/\gamma v$, or large $\xi = b/b_{\max}$.

The large argument asymptotic form of $K_{\nu}(\xi)$ is given in Jackson (3.104)

$$K_{\nu}(\xi) \rightarrow \sqrt{\frac{\pi}{2\xi}} e^{-\xi} \quad (8)$$

for which the energy transfer

$$\Delta E_{\text{adiabatic}} \approx \frac{\pi z^2 e^4}{mv^2} \left(1 + \frac{1}{\gamma^2}\right) \frac{\xi}{b^2} e^{-2\xi} \quad (9)$$

is exponentially small for large ξ .

When $\xi \ll 1$ or $b \ll b_{\max}$, we can use the small argument asymptotic form given in Jackson (3.103)

$$K_0(\xi) \rightarrow -\left(\ln \frac{\xi}{2} + \gamma_{\text{Euler}}\right) \quad K_1(\xi) \rightarrow \frac{1}{\xi} \quad (10)$$

giving an energy transfer

$$\Delta E \approx \frac{2z^2 e^4}{mb^2 v^2} \xi^2 \left[\frac{1}{\gamma^2} \left(\ln \frac{\xi}{2} + \gamma_{\text{Euler}} \right)^2 + \frac{1}{\xi^2} \right] \approx \frac{2z^2 e^4}{mb^2 v^2} \quad (11)$$

This agrees with the result of problem 13.1 where $b \gg b_{\min}$ (but for ξ to be small, we must in the mean time have $b \ll b_{\max}$).